EFFECTIVENESS OF FORMAL OBJECTS WHEN X' -> S IS SEPARATED

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 6236-6240 of sources/stacks-artin/pinned-original-source.tex; proof span is lines 6242-6327, the complete author proof (84 body lines).

Editorial note. Navigation stubs: 8 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 10 of queue_artin.json (pb=3285, stmt 6236-6240, proof 6242-6327).

Editorial note. stacks-artin wave ar01 (2026-10-14): fresh adjudication accept per lane-s3/artin_judgments.md; pool per Chief 10:5x direct dispatch (idle-first over S12's 01:5x note, SMC precedent); artin.tex 99 rows/67 with proof judged 46 accept / 21 reject / 5 AGP-overlap avoidance rows; patch-sd-envs contract spans scanned.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 090T: sources/stacks-artin/pinned-original-source.tex, source lines 6327–6327. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0AQH: sources/stacks-artin/pinned-original-source.tex, source lines 6327–6327. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07X8: sources/stacks-artin/pinned-original-source.tex, source lines 1323–1326. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GI6: sources/stacks-artin/pinned-original-source.tex, source lines 6175–6178. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GDN: sources/stacks-artin/pinned-original-source.tex, source lines 6327–6327. Cited by the retained statement or proof but not part of

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the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GDU: sources/stacks-artin/pinned-original-source.tex, source lines 6327–6327. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GH8: sources/stacks-artin/pinned-original-source.tex, source lines 5138–5153. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0A4Z: sources/stacks-artin/pinned-original-source.tex, source lines 6327–6327. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *In Situation 0GH8 assume $X' \to S$ is separated. Then every formal object for F is effective.*

Proof. A formal object $\xi = (R, \xi_n)$ of F consists of a Noetherian complete local S-algebra R whose residue field is of finite type over S, together with elements $\xi_n \in F(\mathrm{Spec}(R/\mathfrak{m}^n))$ for all n such that $\xi_{n+1}|_{\mathrm{Spec}(R/\mathfrak{m}^n)} = \xi_n$. By the discussion in the proof of Lemma 0GI6 we see that either ξ is a formal object of U' or a formal object of W. In the first case we see that ξ is effective by Lemma 07X8. The second case is the interesting case. Set $V = \mathrm{Spec}(R)$. We will construct an element $(Z, u', \hat{x}) \in F(V)$ whose image in $F(\mathrm{Spec}(R/\mathfrak{m}^n))$ is ξ_n for all $n \geq 1$.

We may view the collection of elements ξ_n as a morphism

$$\xi : \mathrm{Spf}(R) \longrightarrow W$$

of locally Noetherian formal algebraic spaces over S. Observe that ξ is *not* an adic morphism in general. To fix this, let $I \subset R$ be the ideal corresponding to the formal closed subspace

$$\mathrm{Spf}(R) \times_{\xi, W} W_{red} \subset \mathrm{Spf}(R)$$

Note that $I \subset \mathfrak{m}_R$. Set $Z = V(I) \subset V = \mathrm{Spec}(R)$. Since R is $\mathfrak{m}_R$-adically complete it is a fortiori I-adically complete (Algebra, Lemma 090T). Moreover, we claim that for each $n \geq 1$ the morphism

$$\xi|_{\mathrm{Spf}(R/I^n)} : \mathrm{Spf}(R/I^n) \longrightarrow W$$

actually comes from a morphism

$$\xi'_n : \mathrm{Spec}(R/I^n) \longrightarrow W$$

Namely, this follows from writing $W = \mathrm{colim}\, W_n$ as in the proof of Lemma 0GI6, noticing that $\xi|_{\mathrm{Spf}(R/I^n)}$ maps into W_n, and applying Formal Spaces, Lemma 0AQH to algebraize this to a morphism $\mathrm{Spec}(R/I^n) \to W_n$ as desired. Let us denote $\mathrm{Spf}'(R) = V_{/Z}$ the formal spectrum of R endowed with the I-adic topology – equivalently the formal completion of V along Z. Using the morphisms ξ'_n we obtain an adic morphism

$$\hat{x} = (\xi'_n) : \mathrm{Spf}'(R) \longrightarrow W$$

of locally Noetherian formal algebraic spaces over S. Consider the base change

$$\mathrm{Spf}'(R) \times_{\hat{x}, W, g} X'_{/T'} \longrightarrow \mathrm{Spf}'(R)$$

This is a formal modification by Algebraization of Formal Spaces, Lemma 0GDN. Hence by the main theorem on dilatations (Algebraization of Formal Spaces, Theorem 0GDU) we obtain a proper morphism

$$V' \longrightarrow V = \mathrm{Spec}(R)$$

which is an isomorphism over $\mathrm{Spec}(R) \setminus V(I)$ and whose completion recovers the formal modification above, in other words

$$V' \times_{\mathrm{Spec}(R)} \mathrm{Spec}(R/I^n) = \mathrm{Spec}(R/I^n) \times_{\xi'_n, W, g} X'_{/T'}$$

This in particular tells us we have a compatible system of morphisms

$$V' \times_{\mathrm{Spec}(R)} \mathrm{Spec}(R/I^n) \longrightarrow X' \times_S \mathrm{Spec}(R/I^n)$$

Hence by Grothendieck's algebraization theorem (in the form of More on Morphisms of Spaces, Lemma 0A4Z) we obtain a morphism

$$x' : V' \to X'$$

over S recovering the morphisms displayed above. Then finally setting $u' : V \setminus Z \to X'$ the restriction of x' to $V \setminus Z \subset V'$ gives the third component of our desired element $(Z, u', \hat{x}) \in F(V)$. $\square$